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Al-Muneer Hall Reservations by Ahmed Ghaleb- PSM 1-33-44

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CHAPTER 4

REQUIREMENT ANALYSIS AND DESIGN

4.1 Introduction

The requirements and design of the Al-Muneer Online Portal are described in this chapter, along with the description of the system architecture, database structure, principles of user interface design, and required processes. The primary goal of this stage, conducted within PSM1, is the transformation of the requirements of the stakeholders and goals of the project defined in previous stages into a thorough design plan of the system in PSM2. This design is oriented towards providing an intuitive and efficient system to the Visitor, who is interested in information about Al-Muneer Hall, and Owner, who is responsible for the operation of the portal.

4.2 Requirement Analysis

Requirement analysis consists of identifying the user needs as well as the attributes of the system. For the Al-Muneer Online Portal, the requirements were mainly obtained by engaging with the stakeholder, Mr. Almunajid, who is the owner of the Al-Muneer Hall, as discussed in the Case Study (Chapter 2.2.4). Furthermore, a study was performed on the usual features expected of a similar online event management portal for the Yemen environment.

4.2.1 Functional Requirements

Functional requirements describe what the system should do. For the Al-Muneer Online Portal, these are categorized by user roles and key system operations:

FR1: Visitor Functionalities

- **FR1.1 View Venue Viewing:** The system will show complete venue information such as the description, capacity, offered services, and location information.
- **FR1.2 View Media Gallery:** The system will let visitors to view a gallery of images and possibly videos of the venue.

- **FR1.3 Check Availability:** The system will offer an interactive calendar where visitors can see which days have already been booked and which are free.
- **FR1.4 View Pricing Information:** The system will show price information. It could be basic prices, as well as additional charges depending on what services visitors choose.
- **FR1.5 View FAQ:** The system will display a list of Frequently Asked Questions along with their answers.
- **FR1.6 Submit Booking Inquiry:** The system will offer a form for visitors to submit an inquiry, including their name, contact information, date(s) and other specific requests.
- **FR1.7 Submit Payment Proof (Optional):** Submitting Payment Proof (optional): The system will give visitors an option, either before submitting an inquiry or at the admin's request, to submit a picture (a screenshot of a local transfer receipt, for example) as evidence of their payment. The system will accept only common image formats (for instance, JPG, PNG) and set a file size limit.
- **FR1.8 Submit Feedback:** The system shall provide a form for visitors to submit feedback about the venue or their experience with the portal.

FR2: Administrator Functionalities

- **FR2.1 Secure Admin Login:** The system shall require administrators to enter their login information with unique, authenticated credentials (username and hashed password). The system shall display an error message in case of authentication failure.
- **FR2.2 Manage Venue Information:** Manage Venue Information The system shall allow for creation, updating and deleting of text information about the hall (descriptions, services, FAQs, etc.) by the administrator.
- **FR2.3 Manage Media Gallery:** It will enable the administrator to add images and videos to the gallery and remove those already added.
- **FR2.4 Manage Availability Calendar:** It will enable the admin to identify the dates as booked or free in the interactive calendar.
- **FR2.5 Manage Pricing:** The administrator will have the ability to set and change base pricing and customizable pricing options.
- **FR2.6 Manage Booking Inquiries:** This will allow the administrator to manage the booking inquiries that have been sent.
- **FR2.7 Manage Payment Proofs:** It will allow the administrator to manage the payment proofs that are uploaded and change the booking status (like "Payment Verified" or "Pending Confirmation").
- **FR2.8 Feedback Management** The system shall enable the administrator to see the feedback submitted by the user.
- **FR2.9 Generate Basic Reports:** The administrator shall be able to view basic reports such as a list of inquiries in a date range, a summary of booking statuses, or an overview of feedback received.
- **FR2.10 Manage Notifications:** The system will allow the administrator to manage the message templates pre-set for notifications such as inquiry received and payment proof verified, and to

communicate using preferred modes, especially through WhatsApp triggers, while email will be a fallback mode.

FR3: General System Functionalities

- **FR3.1 Responsive Design:** The system shall be responsive and accessible on common web browsers across various devices (desktops, tablets, smartphones).
- **FR3.2 Secure Data Handling:** Safe Data Processing: The system will process all data submitted by the users, which includes personal data from the inquiry and payment proof upload and feedback.
- **FR3.3 Notification Generation:** Based on admin configuration and specific system events (e.g, new inquiry, payment proof submission), the system shall generate notifications to be sent to relevant parties (client or admin).

4.2.2 Non-Functional Requirements

The non-functional requirements define the quality characteristics of the system.

- **NFR1 Performance:** The system shall load pages in an acceptable time frame (e.g., 3-5 seconds on a typical internet connection). The interactive calendar should be responsive to user
- **NFR2 Usability:** The site is easy to use and intuitive for the visitors and administrator. The preferred language will be Arabic with an alternative English language option in case it is feasible to implement under the current scope of the project.
- **NFR3 Security:**
 - **NFR3.1 Admin Authentication:** The admin login shall be protected against common security vulnerabilities. Passwords should be hashed.
 - **NFR3.2 Data Protection:** Users data provided by users and data uploaded by users (payment proofs) should be protected against unauthorized access.
 - **NFR3.3 SSL/TLS:** Client-to-server communication must be encrypted using HTTPS.
- **NFR4 Reliability:** The portal must be generally available for its time, without being unavailable.
- **NFR5 Maintainability:** Code should be well structured and commented and follow the chosen MVC pattern to allow for future updates or modifications (relevant for PSM2 and beyond).
- **NFR6 Scalability (Basic):** NFR6: Not anticipating having a very high number of concurrent users from the get go, the platform architecture (Spring Boot and selected database) need to be scalable to handle a decent amount of increase in visitors (10,000 visitors per month).

4.2.3 Use Case Table and Diagram

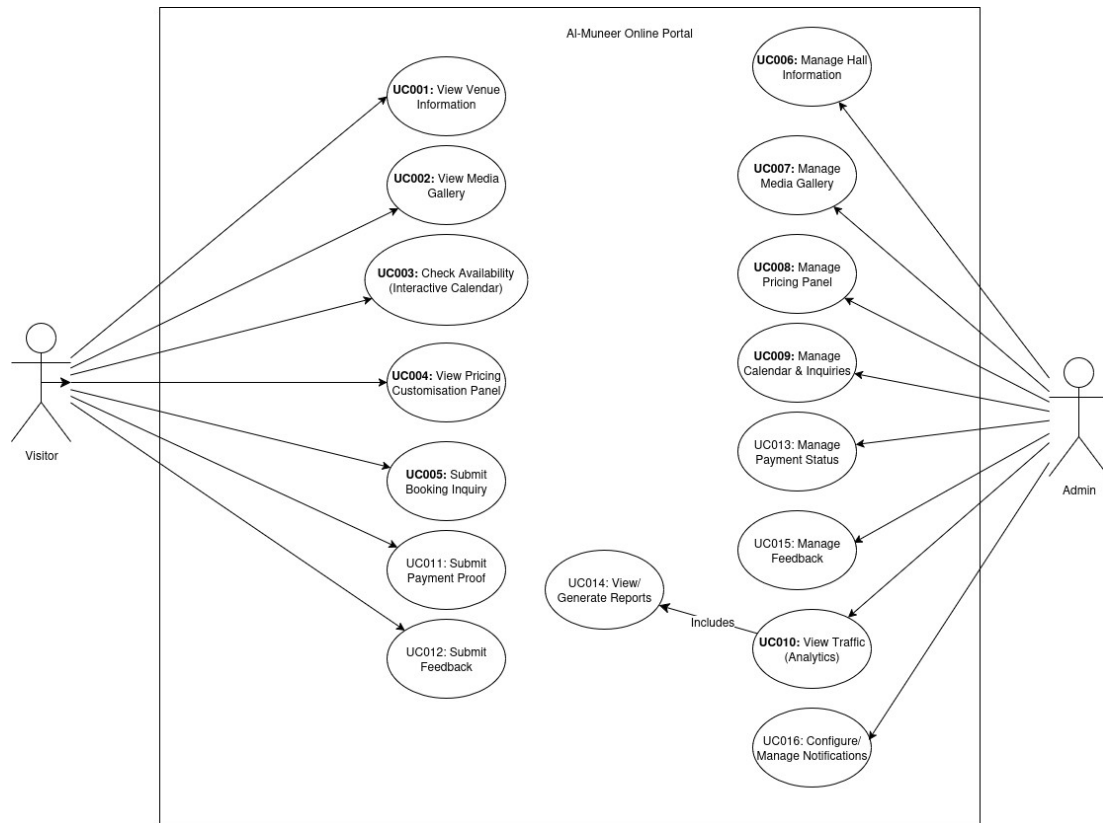


Figure 4.1: Use Case Diagram for the Al-Muneer Online Portal

4.2.3.1 Selected Use Case: Submit Booking Inquiry (UC005)

Table 4.1: Use Case Description for 'Submit Booking Inquiry' (UC005)

Attribute	Description
Use Case ID	UC005
Use Case Name	Submit Booking Inquiry
Actor(s)	Visitor
Pre-condition(s)	Visitor is on the Al-Muneer Online Portal and has navigated to the inquiry section/form.
Normal	1. Visitor is provided with the booking inquiry form.

Flow	2. Visitor enters the necessary information in the specified fields (for example, name, contact, requested date, event type, number of guests, any special requirements). 3. Visitor submits the form. 4. The system verifies the data entered. 5. System saves the inquiry details to the database. 6. System displays a success message to the Visitor. 7. The system sends a notification to the Administrator regarding the inquiry.
Alternative Flow	4a. Invalid Input Data: 4a1. The system shows an error message with the list of fields having invalid data. 4a2. Visitor fixes the data and resubmits (returns to step 3). 7a. Notification Failure: 7a1. The system records the failure of the notification process but the inquiry data is saved successfully. The admin can view the inquiry on logging in.
Post-condition(s)	● Booking inquiry is successfully recorded in the system. ● Administrator is notified of the new inquiry. ● Visitor receives confirmation that their inquiry has been submitted.

4.2.3.2 Sequence Diagram (Submit Booking Inquiry - UC005)

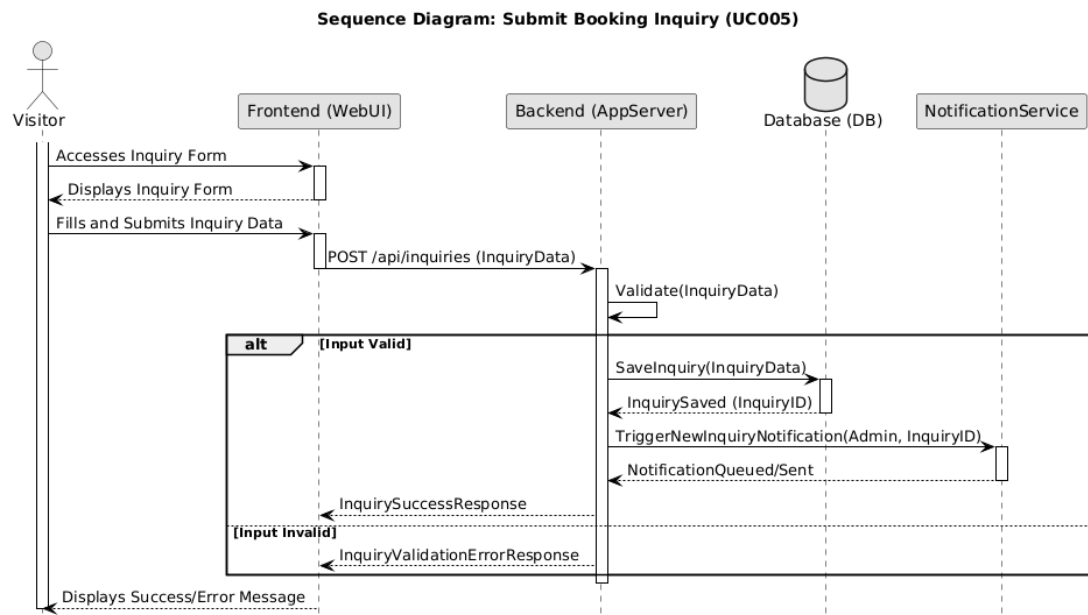


Figure 4.2: Sequence Diagram for 'Submit Booking Inquiry' (UC005)

4.2.3.3 Activity Diagram (Submit Booking Inquiry - UC005)

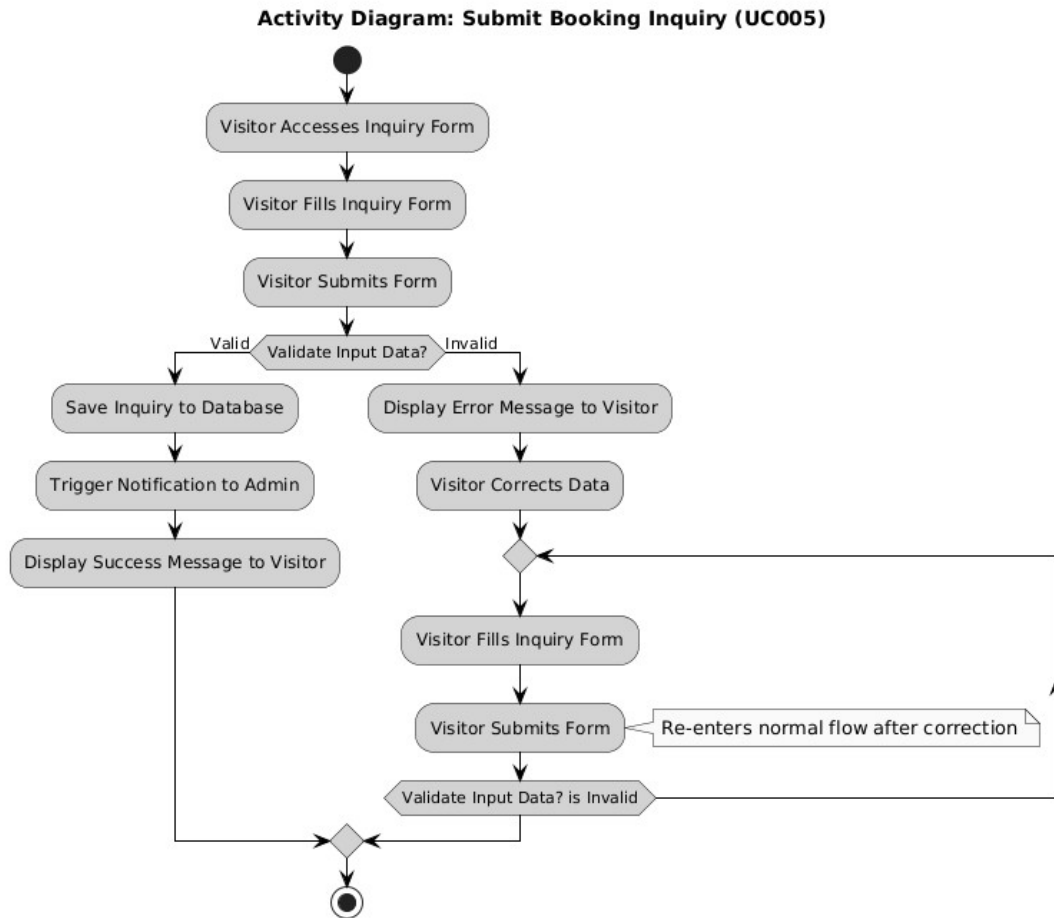


Figure 4.3: Activity Diagram for Submit Booking Inquiry (UC005)

4.3 Project Design

Al-Muneer Online Portal is an MVC Architecture-based platform operating through a single application only. The MVC architecture will ensure that there will be a well-defined division of roles including Model (including data and business logic), View (user interface), and Controller (managing user inputs and application flow). Such approach is extremely suitable for the requirements of the current project.

4.3.1 System Architecture Diagram

The below diagram illustrates the overall MVC architecture employed by the AI-Muneer Online Portal. In this combined system, the Spring Boot framework is used to handle the business logic as well as the User Interface design, creating an integrated and sustainable system.

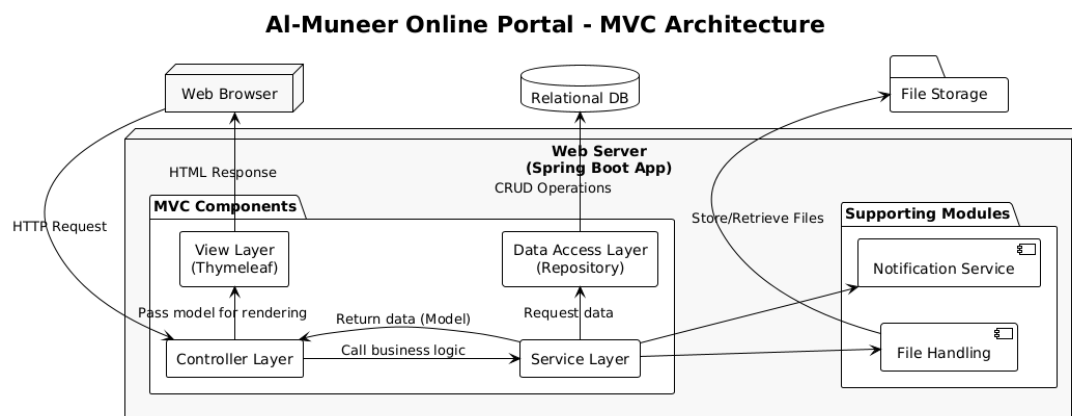


Figure 4.4: System Architecture Diagram (MVC)

4.3.2 Database Design

The reason for choosing the relational database model is due to the organized nature of the data. The following diagram is a basic ERD. The Primary Key (PK) and Foreign Key (FK) will be identified in the detailed design.

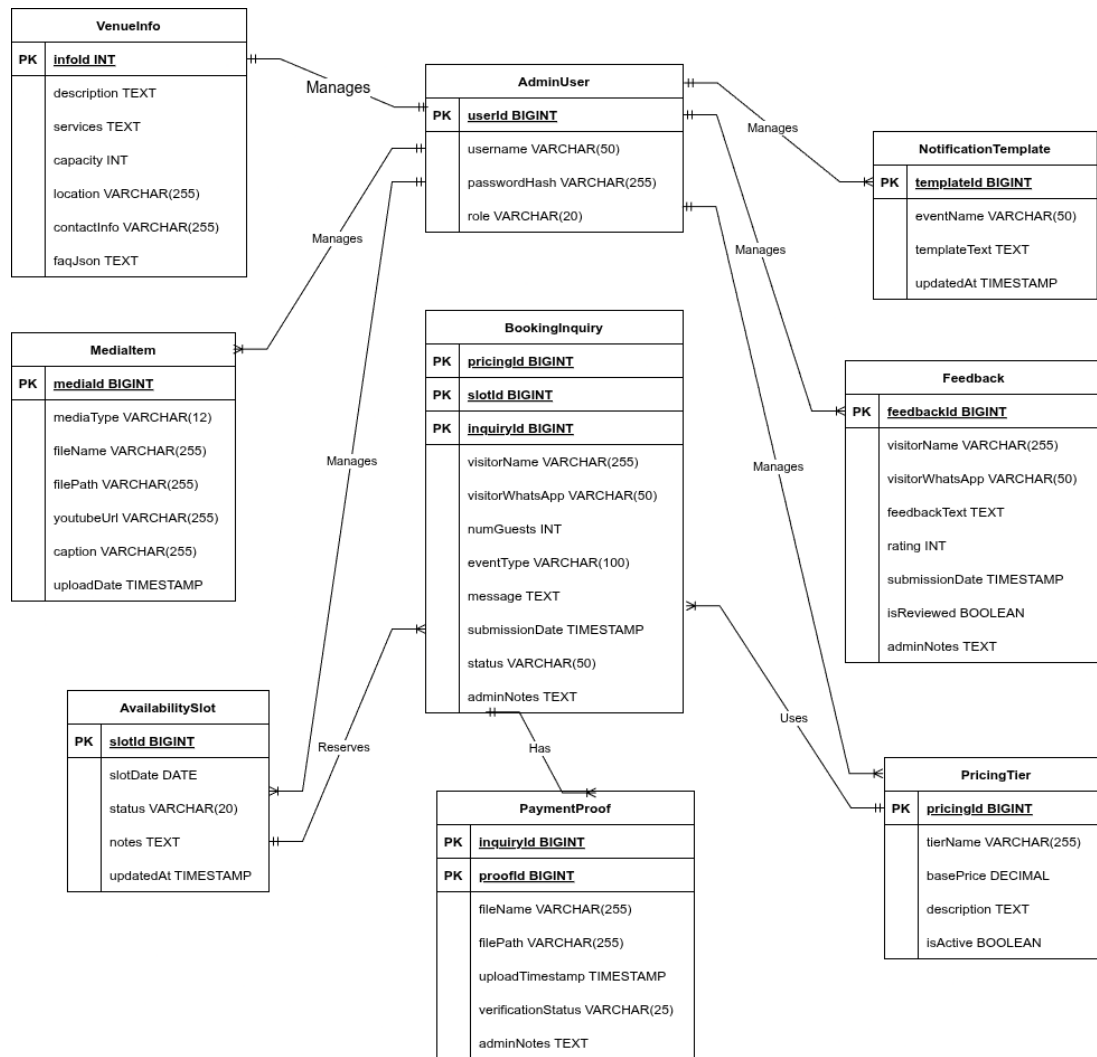


Figure 4.5: Conceptual Entity-Relationship Diagram (ERD)

Data Components:

- AdminUser: Stores admin auth and roles
- VenueInfo: General descriptive information about the venue
- MediaItem: Information regarding uploaded images and videos for the gallery
- AvailabilitySlot: Displays specific dates and whether they are booked or free.
- PricingTier : Details of various pricing packages or rates.
- BookingInquiry: Data about the inquiries of visitors.
- PaymentProof: Metadata of uploaded payment proof files, linked to inquiries.
- Feedback: Collects user submitted feedback.

4.4 Interface Design (Low-Fidelity)

4.4.1 Home Page

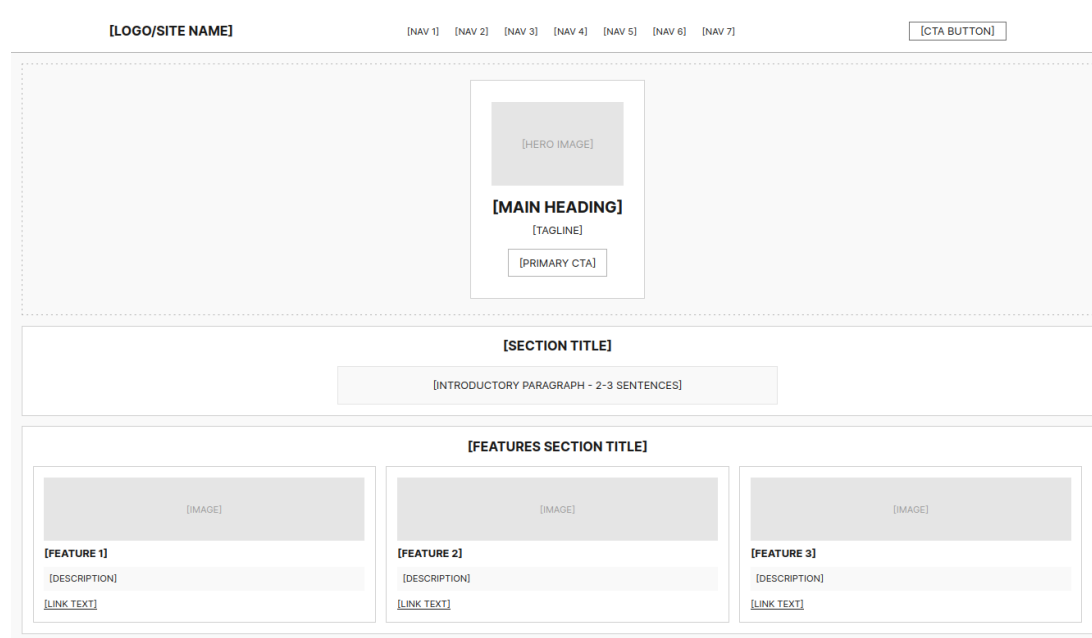


Figure 4.6: Low-Fidelity Wireframe: Homepage

4.4.2 Photo Gallery

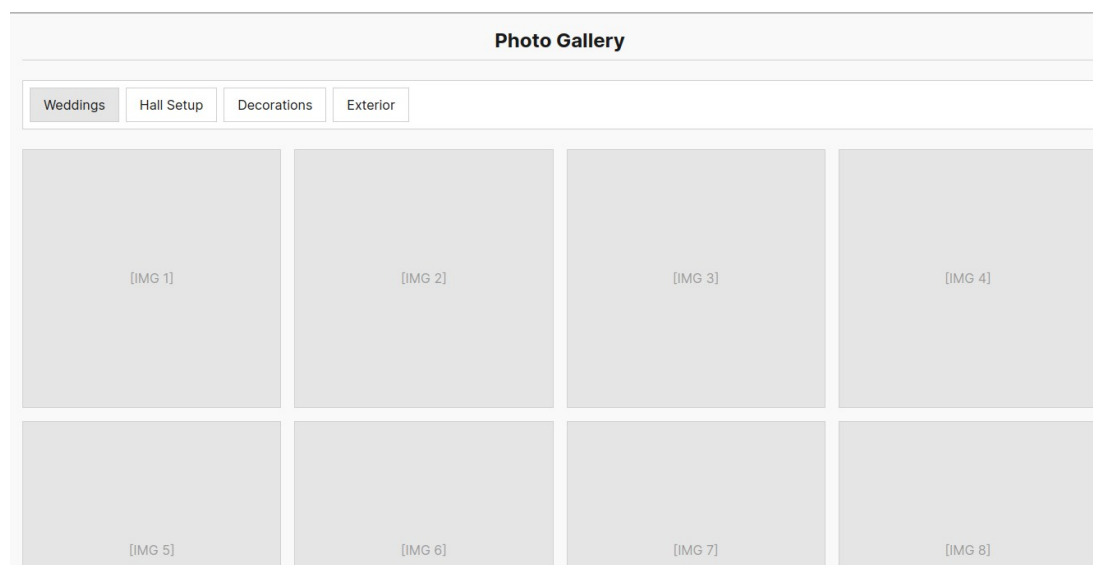


Figure 4.7: Low-Fidelity Wireframe: Gallery Page

4.4.3 Availability calendar

Check Availability

<
May 2025
>

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

☐ Available
 ☐ Pending
 ☐ Booked

Make an Inquiry

Figure 4.8: Low-Fidelity Wireframe: Availability Calendar Page

4.4.4 Pricing Page

[PAGE TITLE]

[INTRODUCTORY TEXT BLOCK]

[PACKAGE 1]
[PRICE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]

[BUTTON TEXT]

[PACKAGE 2]
[PRICE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]

[BUTTON TEXT]

[PACKAGE 3]
[PRICE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]
 ✓ [FEATURE]

[BUTTON TEXT]

[SECTION TITLE]

[SERVICE NAME]
[PRICE INFO]

[SERVICE NAME]
[PRICE INFO]

[SERVICE NAME]
[PRICE INFO]

[SERVICE NAME]
[PRICE INFO]

[SERVICE NAME]
[PRICE INFO]

[SERVICE NAME]
[PRICE INFO]

[CTA BUTTON]

Figure 4.9: Low-Fidelity Wireframe: Pricing Page

4.4.5 Inquiry Form Page

[PAGE TITLE]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
mm / dd / yyyy

[FIELD LABEL]
[SELECT OPTION]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[PLACEHOLDER]

[FIELD LABEL]
[SELECT OPTION]

[PRIVACY/DISCLAIMER TEXT]

[SUBMIT BUTTON]

[SIDEBAR TITLE]

[CONTACT METHOD]: [CONTACT INFO]

[CONTACT METHOD]: [CONTACT INFO]

[INFO TITLE]:
[INFO LINE 1]
[INFO LINE 2]

Figure 4.10: Low-Fidelity Wireframe: Inquiry Form Page